A complete $n = 3$ answer for the K_nPK_n extreme-ray problem

Solution packet for arXiv:2006.14543

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Status. `partial_solution_likely_valid`. This packet completely settles the first omitted dimension $n = 3$ of an all-dimensions open problem in Mueller-Hermes [1]. The proof is an exact finite polyhedral enumeration; it uses no floating-point decisions. The general problem for arbitrary n remains open.

Source question

The question occurs in Section 9, "Conclusion and open questions," on printed page 38 of [1]. Here E_n is the all-ones matrix and "ew-positive" means entrywise nonnegative.

- For every $n \in \mathbb{N}$ we can consider the symmetric orthogonal matrices

$$K_n = \frac{2}{n}E_n - \mathbb{1}_n.$$

What are the ew-positive matrices $P \in \mathcal{M}_n(\mathbb{R})$ such that K_nPK_n is ew-positive as well? How many extremal rays does the corresponding polyhedral cone have?

The relevant bullet asks: for

$$K_n = \frac{2}{n}E_n - I_n,$$

which entrywise-nonnegative $P \in M_n(\mathbb{R})$ also have K_nPK_n entrywise nonnegative, and how many extreme rays does the resulting polyhedral cone have?

Result at a glance

Put

$$K = K_3 = \frac{2}{3}E_3 - I_3, \qquad \mathcal{C}_3 = \{P \in M_3(\mathbb{R}) : P \geq_{\mathrm{ew}} 0,\ KPK \geq_{\mathrm{ew}} 0\}.$$

Let G be the symmetry group generated by independent permutations of rows and columns, transposition, and the involution $P \mapsto KPK$. Then $\mathcal{C}_3$ has exactly 78 extreme rays, arranged into three G-orbits. Representatives and their images under the involution are

$$P_A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{pmatrix}, \qquad KP_AK = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$P_B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 2 \end{pmatrix}, \qquad KP_BK = \begin{pmatrix} 2 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$P_C = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad KP_CK = P_C.$$

The corresponding orbit sizes are 36, 36, and 6.

Proof Intuition

After vectorizing 3×3 matrices, $\mathcal{C}_3$ is a full-dimensional pointed cone in $\mathbb{R}^9$ cut out by only 18 rational inequalities: nine coordinates of P and nine coordinates of KPK. A ray of a nine-dimensional polyhedral cone is extreme exactly when its active inequalities have rank eight. Thus every extreme ray is exposed by some choice of eight among the 18 inequalities.

This turns exhaustiveness into a small finite proof. For each of the $\binom{18}{8} = 43{,}758$ choices, compute the exact rational nullspace. A one-dimensional nullspace contributes a candidate precisely when one of its two orientations satisfies all 18 inequalities. Primitive-integer normalization removes duplicates. The resulting 78 vectors all have active rank eight. Applying the natural symmetries produces exactly the three displayed orbits. What remains open in the source is the uniform classification for general n, not the $n = 3$ case established here.

Formal theorem and proof

We flatten matrices row by row. Since $K = K^T$, this convention gives

$$\operatorname{vec}(KPK) = (K \otimes K) \operatorname{vec}(P).$$

Set

$$T = K \otimes K, \qquad A = \begin{pmatrix} I_9 \\ T \end{pmatrix} \in M_{18\times 9}(\mathbb{R}).$$

Thus $\operatorname{vec}(\mathcal{C}_3) = \{x \in \mathbb{R}^9 : Ax \geq 0\}$.

Lemma 1 (Active-rank criterion)**.** *Let $D = \{x \in \mathbb{R}^d : Bx \geq 0\}$ be a full-dimensional pointed polyhedral cone. For nonzero $x \in D$, let $B_0(x)$ consist of the rows b of B with $bx = 0$. Then $\mathbb{R}_+ x$ is an extreme ray of D if and only if* $\operatorname{rank} B_0(x) = d - 1$.

Proof. The minimal face of D containing x is obtained by imposing precisely the equalities active at x. Its linear span is $\ker B_0(x)$, hence its dimension is $d - \operatorname{rank} B_0(x)$. The face is a ray exactly when this dimension is one. Pointedness excludes a one-dimensional linear subspace. □

Theorem 1. *The cone $\mathcal{C}_3$ has exactly* 78 *extreme rays. Under G they are the three orbits of P_A, P_B, P_C displayed above, of respective sizes* $36, 36, 6$.

Proof. First, $K^2 = I_3$ and $K\mathbf{1} = \mathbf{1}$. The all-ones matrix is therefore sent to itself and satisfies all 18 inequalities strictly, so $\mathcal{C}_3$ is full-dimensional. It is pointed because $P \geq_{\mathrm{ew}} 0$ and $-P \geq_{\mathrm{ew}} 0$ force $P = 0$.

We now give an exhaustive finite enumeration. For every eight-element subset $I \subset \{1, \ldots, 18\}$, form the 8×9 submatrix A_I. If $\ker A_I$ is one-dimensional, choose a nonzero generator x_I. Retain its ray if either $Ax_I \geq 0$ or $A(-x_I) \geq 0$, and normalize the retained orientation to a primitive integer vector. Deduplication gives 78 rays.

This procedure is exhaustive: by Lemma 1, every extreme ray has eight linearly independent active rows, hence occurs for at least one of the subsets I. Conversely, every retained nonzero vector has at least the eight independent active rows in I; its full active matrix cannot have rank nine, because it annihilates that nonzero vector. Its active rank is therefore eight, so Lemma 1 shows that every retained ray is extreme.

All arithmetic in this enumeration is rational. The verifier `code/verify_n3_extreme_rays.py` implements exactly the preceding loop using SymPy nullspaces. It tests all 43,758 subsets, reports 78 primitive rays, and then generates G using row and column permutations, transposition, and T. The rays split into disjoint orbits represented by P_A, P_B, P_C, with sizes $36, 36, 6$. Their ranks are respectively $1, 2, 3$, so the three orbits are distinct; rank is preserved by every generator of G. Direct exact multiplication gives the three displayed matrices KP_iK. The active-constraint ranks reported for all three representatives are eight. Since $36 + 36 + 6 = 78$, these orbits exhaust the list. □

Verification report

Run, from the packet directory,

```
conda run --no-capture-output -n sandbox python \
  code/verify_n3_extreme_rays.py
```

The expected summary is

```
active_bases_tested=43758
extreme_rays=78
symmetry_orbits=3
orbit sizes: 36, 36, 6
active ranks: 8, 8, 8
```

The computation constitutes a proof because it exhausts every possible rank-eight active basis and all nullspace, feasibility, normalization, rank, and orbit operations are exact over $\mathbb{Q}$. It is not a sampling or floating-point polyhedral calculation. The short script is included so that each finite step can be independently audited.

As hand checks, $K^2 = I$, each displayed P_i and KP_iK is entrywise nonnegative, and the three support-size pairs are $(4, 4), (4, 4), (3, 3)$. The verifier reports respectively $10, 10, 12$ active inequalities and active rank eight in every case.

Scope, novelty, and human review

Scope. The theorem completely answers the source's matrix-cone question only for $n = 3$. It does not classify arbitrary n, the Pauli PPT cone for $N \geq 3$, or the separate tensor-power questions on the same source page.

Novelty check. The run indexes had no result for arXiv:2006.14543 or this cone. A bounded current search used the exact source wording, the expressions K_nPK_n and "Pauli PPT spectra," the paper title and author, and later arXiv papers and citations concerning its extremal-ray problem. It found the source and general later work on positive-map tensor powers, but no $n = 3$ classification or general answer to this exact matrix-cone question. Novelty confidence is moderate because the search is bounded and the result is an elementary finite polyhedral classification.

Human-review recommendation. Review as an exact computer-assisted partial solution. The principal checks are (i) the row-major vectorization identity giving $T = K \otimes K$, (ii) the active-rank criterion, and (iii) that the verifier truly iterates all 43,758 eight-row subsets using exact arithmetic. A secondary check is whether the source authors intended n to range over all positive integers or only the multi-qubit subsequence; the printed question explicitly says every $n \in \mathbb{N}$, so $n = 3$ is within its literal scope.

References

[1] A. Mueller-Hermes, *Decomposable Pauli diagonal maps and Tensor Squares of Qubit Maps*, arXiv:2006.14543; Journal of Mathematical Physics **62** (2021), 092204.